Midterm 2 Review

ECON 441: Introduction to Mathematical Economics

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What's on the exam

- Wednesday, October 21, in class. 60 minutes, 20 points.
- Module III: Calculus (Lectures 8–12; integration is not covered)
- A sample exam with solutions and the help sheet provided with the exam are on the Exams page
- Closed book: no phones or computers
- For your review: lecture slides and worksheets, Practice Problems 8–12 (also collected as one PDF for the module), the module notes, and additional problems in the textbook

The Derivative

Average rate of change, then let the change shrink:

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{f\,(x + \Delta x) - f\,(x)}{\Delta x}$$

Use the limit definition to find $f'(x)$ for $y = f(x) = x^2 + 3x$.

Limits and Continuity

$\lim_{x \to a} f(x) = L$ exists only if the left-side and right-side limits agree.

f is **continuous** at a if the limit exists and $\lim_{x \to a} f(x) = f(a)$.

$$f(x) = \begin{cases} x + 1 & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

Does $\lim_{x \to 1} f(x)$ exist? Is f continuous at $x = 1$?

Continuity and Differentiability

Continuity is **necessary** for differentiability, but not **sufficient**.

$$g(x) = \begin{cases} 2x & \text{if } x < 1 \\ x + 1 & \text{if } x \geq 1 \end{cases}$$

Plot g. Is it continuous at $x = 1$? Is it differentiable there?

The Rules

Constant, power, sum-difference, and the inverse-function rule $\frac{dx}{dy} = \frac{1}{dy/dx}$; then

$$\frac{d}{dx}[fg] = fg' + f'g \qquad \frac{d}{dx}\frac{f}{g} = \frac{f'g - fg'}{g^2} \qquad \frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

Find dy/dx:

- $y = (x^2 + 1)(3x - 2)$
- $y = \frac{x}{x + 2}$
- $y = (2x + 1)^3$

Exponential and Log Functions

$$\frac{d}{dx}e^{f(x)} = f'(x)e^{f(x)} \qquad \frac{d}{dx}\ln f(x) = \frac{f'(x)}{f(x)}$$

$$\ln(uv) = \ln u + \ln v \qquad \ln(u/v) = \ln u - \ln v \qquad \ln u^a = a \ln u$$

Find dy/dx:

- $y = 3e^{2x}$
- $y = x^2 \ln x$
- $y = \ln(x^2 + 4)$

Elasticity in Logs

$$\varepsilon = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}$$

Demand satisfies $\ln Q = 2 - 0.5 \ln p$. What is the price elasticity of demand? Is demand elastic or inelastic?

Partial Derivatives

Differentiate with respect to one variable, holding the others constant.

$$f(x, y) = x^2 y + y^3$$

Find f_x, f_y, and the gradient ∇f. Is f_x increasing in y?

Production Function

$$Q = K^{0.3} L^{0.7}$$

Find the marginal product of capital Q_K and of labor Q_L.

Is the marginal product of labor decreasing in L?

Elasticity

$$\varepsilon = \frac{\text{Percentage change in } y}{\text{Percentage change in } x} = \frac{dy}{dx} \cdot \frac{x}{y}$$

Demand is $Q = 50 - 2p$. Find the elasticity at $p = 10$. Elastic or inelastic?

Total Differential and Total Derivative

For $y = f(x_1, x_2)$:

$$dy = f_1\, dx_1 + f_2\, dx_2 \qquad \frac{dy}{dx_1} = f_1 + f_2 \frac{dx_2}{dx_1}$$

$z = x^2 + xy$, where $x = t$ and $y = t^2$. Find dz and dz/dt.

Total Derivative with Time

Output $Y(t) = K(t) + N(t)$, where $K(t) = 3t$ and $N(t) = e^{0.5t}$.

Find the rate of change of output over time, dY/dt.

Implicit-Function Rule

For $F(x, y) = 0$, totally differentiate: $F_x\, dx + F_y\, dy = 0$, so

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

$F(x, y) = x^2 + y^2 - 25 = 0$. Find dy/dx, and its value at $(3, 4)$.

Market Equilibrium

Demand $Q^d = 60 - 3P + 0.1Y$, supply $Q^s = 10 + 2P$.

Write the equilibrium condition as $F(P, Y) = 0$ and find $\partial P/\partial Y$ by the implicit-function rule. Check by solving for P.